

HEIMDALL: Smart Campus Access & Hostel Management

Infrastructure Cost Feasibility & Zero-Cost Operating Report | 5,000 Active Campus Users

EXECUTIVE SUMMARY: HEIMDALL can be operated for 100% FREE (\$0.00 / month) for 5,000 students. By pairing Cloudflare R2 Automated Archival with optimized resource usage, database growth is permanently capped, eliminating expensive cloud infrastructure bills.

1. DATABASE: Why MongoDB Atlas Free Tier (\$0/mo) Works

Why \$0 (M0 Free Tier) Works:

- **Storage Safety (512 MB Limit):** 5,000 students generate ~150 MB of raw logs per month. By setting the automated R2 archival retention to **1 month** (`ARCHIVE_RETENTION_MONTHS=1`), logs older than 30 days are compressed, uploaded to Cloudflare R2, and deleted from MongoDB. Live database storage stays capped at **~100 MB – 150 MB** forever.
- **Connection Safety (500 Max Connections):** Node.js uses Mongoose Connection Pooling (10–15 total connections). Whether 10 or 5,000 students access the portal, the backend maintains only ~15 connections to MongoDB, remaining well under the 500 limit.

Why \$9/month (M2 Tier) Was Mentioned Originally:

- The \$9/mo M2 tier provides 2 GB storage and higher throughput. It was listed as an optional upgrade only for heavy concurrent gate rush scenarios (e.g., 500+ students scanning within 10 minutes). However, it is **NOT required to start**.

2. BACKEND SERVER: Why \$15–\$25 Was Listed & 3 Ways to Get \$0/mo

Why \$15–\$25/month Was Listed Originally:

Free hosting services (e.g., Render.com) put free servers to 'sleep' after 15 minutes of inactivity. When asleep, a student's QR scan triggers a **20–30 second delay ('cold start')** to wake up the server. A paid VPS (\$15–\$25/mo) stays awake 24/7/365 for instant < 1-second scans.

3 Ways to Run the Backend for 100% FREE (\$0/mo) with ZERO Cold Start Delay:

1. **College IT Lab PC / Server (Best Choice):** Deploy Node.js on a spare desktop or server in the campus IT lab connected to campus WiFi/LAN. Fast local scanning, zero internet latency, **\$0.00 / month**.
2. **Oracle Cloud 'Always Free' VPS:** Oracle Cloud provides 2 free cloud servers with 4 vCPUs and 24 GB RAM that run 24/7/365 and **NEVER go to sleep. \$0.00 / month forever**.
3. **Render Free Tier + Free Keepalive Ping:** Deploy on Render.com (Free Tier) and use UptimeRobot.com (Free service) to ping `/api/health` every 10 minutes. This prevents the server from sleeping. **\$0.00 / month**.

3. LONG-TERM ARCHIVES & FRONTEND (\$0.00 / month)

- **Cloudflare R2 (Archive Storage):** 10 GB free monthly storage, 1M free write operations, and **\$0.00 egress (download) fees**. 5,000 students generate ~0.42 GB/year of compressed archives, fitting in the free tier for **over 23 years**.
- **Cloudflare Pages / Vercel (Frontend):** Hosts the React static web application with global CDN distribution at **\$0.00 / month**.

4. 5,000-User Complete Financial Summary Table

Component	Zero-Cost Setup (\$0/mo)	Technical Justification
Database	\$0.00 / mo (MongoDB Atlas M0)	1-month R2 archival keeps DB capped at ~150 MB (under 512MB limit). Connection pool = ~15 conns.
Archive Storage	\$0.00 / mo (Cloudflare R2)	10 GB free tier. 5,000 users generate ~0.42 GB/yr of gzipped JSON. \$0 egress fees.
Backend API	\$0.00 / mo (Campus PC / Oracle Free)	Oracle Free VPS (4 vCPU, 24GB RAM) or College PC runs 24/7 with zero cold starts.
Queue / Cache	\$0.00 / mo (Upstash Redis)	In-memory QR validation and BullMQ queue runs under 10k free daily commands.
Frontend Web	\$0.00 / mo (Cloudflare Pages)	Static React app CDN hosting with unlimited free bandwidth.
TOTAL COST	\$0.00 / month (\$0 / year)	100% Free Operational Infrastructure for 5,000 Students

5. SUMMARY VERDICT & RECOMMENDATION

Deploy HEIMDALL using the **\$0.00 / month Zero-Cost Model**. Set ARCHIVE_RETENTION_MONTHS=1 in the backend environment, use Cloudflare R2 for long-term storage, and host the Node.js server on Oracle Cloud Free VPS or a campus IT lab computer. The platform will operate smoothly, rapidly, and securely at zero financial cost to the institution.